Assignment Report

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1 1. Objective and Context

The objective of this assignment was to expand the design by integrating a suitable body temperature sensor into the battery-free Densor project. Initially, my first candidate was the Sensirion SHT4x family of temperature and humidity sensors, as I already have experience with these sensors and have developed multiple I^2C drivers for different platforms. However, although the SHT4x provides both temperature and humidity measurements, its accuracy is not ideal for precise body temperature monitoring. Moreover, while humidity could potentially provide useful information for the project, measuring air humidity inside the mouth is not necessarily the most appropriate approach. A more suitable solution for oral wetness would be a dedicated wetness sensor, for example based on electrodes.

Based on these considerations, I conducted further research and identified the TI TMP117 and ADI MAX30205 as two high precision temperature sensors that could also meet the requirements of an ASTM E1112 compliant system. After further comparison, I selected the TMP117, as it offers lower power consumption and comes in a smaller package while still providing up to $\pm 0.1\,^{\circ}\mathrm{C}$ accuracy over the operating temperature range of this application.

2 Schematic and PCB integration

For the integration, the TMP117 was added to the schematic with a 100 nF decoupling capacitor and connected to SCL, SDA, and V_{MCU}. The interrupt pin was left floating, as it is not required by the firmware. The same form factor was maintained for the PCB layout. Both the schematic and the PCB layout were designed following the recommendations provided in the TI datasheet.

3 Firmware integration

To ensure a modular approach, a dedicated driver for the TMP117 was developed on top of the STM32 HAL, which was already included in the original firmware. Since the MCU has limited memory resources, the driver was kept barebones, containing only the necessary macros and the implementations of the `readReg()` and `writeReg()` functions. The firmware was extended in the following ways:

- A bit in the `SensorEnable` bitmap of the NFC tag was reserved for the TMP117, and the corresponding firmware checks were updated accordingly.
- The sensor initialization was added. The TMP117 is kept in shutdown mode and switched to one-shot mode only when a temperature measurement is required for the periodic data transaction. Polling was preferred over the interrupt line due to the short conversion time, while also avoiding the need to include the sleep mode HAL layers in the firmware.

Additionally, while this has not been actively done, the sensor can be easily calibrated to compensate for unforeseen noise after testing. Lastly, although the code could not be tested due to the unavailability of the hardware, it was fully documented and a Doxygen configuration file was added to generate the corresponding HTML documentation, the added firmware modifications are in `/* ASSIGNMENT CODE BEGIN */` areas.

4 AI usage disclaimer

LLM models were used to get an initial idea of the schematic organization and for proofreading the grammar of this report. All chat histories can be found in the Extra folder.